Identification of ferroptosis-related hub genes and their potential relation with apoptosis in Ovarian Cancer

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DECLARATION

I hereby declare that the dissertation entitled " **Identification of ferroptosis-related hub genes and their potential relation with apoptosis in Ovarian Cancer**" is submitted to Pondicherry University in partial fulfilment for the award of Master of Science in Bioinformatics is a record of original work carried out by me under the guidance and supervision of **Dr. Basant K. Tiwary**, Professor, Department of Bioinformatics, School of Life Science, Pondicherry University, Puducherry. This work has not been submitted elsewhere for any other degree, diploma, associateship or fellowship.

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ABSTRACT

Ovarian cancer has emerged as a formidable challenge in oncology due to its late diagnosis and high mortality rates. Recent research has demonstrated the complex interactions between ferroptosis and apoptosis, two distinct cell death pathways, in the context of ovarian cancer. Ferroptosis is an iron-dependent form of regulated cell death characterized by lipid peroxidation, has emerged as a promising therapeutic target in various cancers, including ovarian cancer. In contrast, apoptosis is a cell death mechanism involving caspase activation and DNA fragmentation, has been a major focus of cancer therapy for decades.

This study explores the relationship between ferroptosis and apoptosis in ovarian cancer from the data collected from co-expression profile, PPI network, co-expression network and functional gene enrichment analysis. Differential gene expression showed that there were a total of 7,862 genes differentially expressed between healthy samples and ovarian cancerous samples. As a result we retrieved two crucial ferroptosis genes-CDNK1A and GDF, co-expressed with apoptosis genes. Functional enrichment analysis of these two genes revealed theor association with carcinogenic pathway.

In conclusion, understanding the relationship between ferroptosis and apoptosis in ovarian cancer provides insightful observations into the development of innovative therapeutic approaches. Further research is required to unravel the details about this interaction and translate these findings into clinical practice for more effective treatment of ovarian cancer.

1. INTRODUCTION

1.1 Overview of Ovarian Cancer

Ovarian cancer is a complex and often deadly disease that arises from the abnormal growth of cells in the ovaries. This type of cancer is known for its silent and nonspecific symptoms, which can make early detection challenging. Risk factors for developing ovarian cancer include age, family history, genetic mutations, and hormonal factors. There are different subtypes of ovarian cancer, with epithelial ovarian cancer being the most common. Diagnosis typically involves imaging tests, blood tests, and biopsy.[1] Treatment options for ovarian cancer include surgery, chemotherapy, and targeted therapy. Despite advances in research and treatment, ovarian cancer remains a significant health concern for women worldwide, with a high mortality rate. Understanding the overview of ovarian cancer is crucial for improving early detection, treatment, and outcomes for patients.[2]

1.2 Risk Factors Associated with Ovarian Cancer

Ovarian cancer is a complex disease with various risk factors that significantly impact its development. Research indicates that factors such as high gravidity, hysterectomy, female sterilization, and oral contraceptive use may be associated with a reduced risk of ovarian cancer, highlighting the importance of reproductive and contraceptive history in understanding this malignancy (Beral et al., 1989).[3] Additionally, the onset of ovarian cancer seems to be influenced by menopausal events, particularly early menopause, where the mechanisms leading to cancer cell implantation in the ovary are still not fully elucidated. Inflammation, triggered by conditions like ovulatory wound repair, incessant ovulation, ovarian surface scarring, and aging, has been identified as a key factor contributing to the increased risk of ovarian cancer, shedding light on the role of the immune response in disease progression (Jia et al., 2018).[4] These insights underscore the importance of considering both reproductive factors and inflammatory processes in exploring the multifaceted nature of ovarian cancer risk factors.

1.3 Symptoms and Diagnosis of Ovarian Cancer

Symptoms play a crucial role in the diagnosis of ovarian cancer, with studies indicating that early identification of symptoms could potentially lead to a timelier diagnosis. Research suggests that common symptoms such as abdominal pain, urogenital problems, abdominal distension, and changes in bowel habits are often reported by women before a definite diagnosis of ovarian cancer is made. The time between the onset of these symptoms and a confirmed diagnosis can vary, emphasizing the need for efficient and accurate diagnostic strategies.[5] In exploring presenting symptoms and delays in diagnosis, it becomes evident that recorded symptoms in electronic patient records may differ from self-reported data but still provide valuable insights into the diagnostic process. This data underscores the importance of further investigations into symptom lead time and the impact of recording practices on accurate diagnosis in ovarian cancer cases, ultimately highlighting the complexities involved in early detection and management of this disease.

1.4 Treatment Options for Ovarian Cancer

Ovarian cancer poses significant challenges due to its late-stage diagnosis and limited efficacy of standard treatment options such as surgery and chemotherapy. While immunotherapy shows promise as a potential strategy to overcome treatment failure in recurrent cases, the current clinical trials have yet to demonstrate significant clinical benefits (BAZAN et al., 2014). [6] In Belgium, advancements in personalized medicine have revolutionized the approach to treating ovarian cancer, focusing on histological and molecular characteristics to tailor therapy for each patient (Baurain et al., 2020).[7] Individualized approach based on molecular profiling offers hope for more effective and precise treatment strategies in the management of this challenging disease.

1.5 Ongoing Research and Future Directions in Ovarian Cancer

Ongoing research in ovarian cancer is pivotal for advancing treatment strategies and improving patient outcomes. In recent years, however, significant advancements have been made to understand the molecular mechanisms underlying cancer progression, thus leading to the development of novel therapeutic approaches targeting specific cellular pathways.[8] Among these, apoptosis and ferroptosis have emerged as promising mechanism for cancer therapy.

1.6 Apoptosis

Apoptosis, or programmed cell death, is a highly regulated process required for maintaining tissue homeostasis and eliminating damaged or abnormal cells. Dysregulation of apoptosis is a hallmark of cancer, thus allowing malignant cells to avoid death signals and proliferate uncontrollably.[9] Hence, using the apoptotic pathway has become an innovative point in cancer research, with the aim of inducing selective cell death in tumor cells meanwhile sparing healthy tissues.

1.6.1 Role of Apoptosis in Ovarian Cancer Therapy

By targeting malignant cancer cells for destruction apoptosis plays a role in ovarian cancer treatment. However,in ovarian cancer, the ability of the tumor cells to resist apoptosis contributes to chemo-resistance and disease progression. Apoptosis in ovarian cancer involves a complex working of molecular pathways, including the activation of caspases and the regulation of death receptor ligands like TRAIL, TNF, and CD95L. The delicate balance between pro-survival and pro-apoptotic molecules in the ovary influences follicular survival, atresia, and luteogenesis.[10]

Drugs targeting apoptosis, like TRAIL (tumor necrosis factor-related apoptosis-inducing ligand), triggers apoptosis specifically in tumor cells without affecting the healthy cells. By combining chemotherapy along with agents inducing apoptosis, tumor cell death can be increased, even in drug-resistant ovarian carcinoma cells.[11] This approach can be used to overcome the resistance of the tumor cells to apoptosis induced by chemotherapeutic drugs. Understanding these molecular pathways is required to develop effective treatments targeting apoptosis in ovarian cancer cells as targeting apoptosis is one of the most successful non-surgical approaches to cancer therapy.

Many anticancer drugs work by inducing apoptosis in cancer cells through various mechanisms, such as activating pro-apoptotic proteins, inhibiting anti-apoptotic proteins, or triggering death receptor signaling.

For example, drugs that target the BCL-2 family of apoptosis regulators, or inhibitors of apoptosis (IAP) proteins, can help restore the balance of pro- and anti-apoptotic signals in cancer cells and promote their death.

Additionally, the immune system can induce apoptosis in cancer cells through mechanisms like cytotoxic T-cell and NK cell-mediated granzyme release.

Key benefits of targeting apoptosis in cancer treatment are:

1) Eliminating cancer cells, as by inducing apoptosis, or programmed cell death as become an effective way to eliminate cancer cells that have defects in their apoptosis pathways, allowing them to grow uncontrollably.

2) Since many cancer therapies work by triggering apoptosis in cancer cells, resistance to other cancer treatments can be overcome by targeting the defects in the apoptosis pathways

3) Apoptosis is preferred over other mechanisms of cell death for cancer therapy as it causes minimal inflammation and damage to surrounding tissues.

4) By combining apoptosis-inducing agents with other cancer drugs improves the overall efficacy of treatment due to synergistic effects.
5) Drugs targeting important apoptosis regulators like the BCL-2 family proteins and inhibitor of apoptosis (IAP) proteins have shown promise in cancer therapy.

1.6.2 Limitations to use of Apoptosis as Therapy

1) The apoptosis pathways involve a complex network of signaling molecules and interactions. Thus precise targeting of the correct points in these pathways in order to induce cancer cell death can prove to be difficult.

2) It is difficult to completely avoid off-target effects and damage to healthy cells as targeting apoptosis is more specific to cancer cells as compared to other traditional treatments

3) Cancer cells can develop resistance to apoptosis-inducing therapies by various mechanisms, for example such as mutations in key apoptosis regulators like p53 or up-regulation of anti-apoptotic proteins. Overcoming these resistance mechanisms can become a difficult task

4) Certain apoptosis-targeting agents, like IAP inhibitors, can have acute or chronic normal tissue toxicities that need to be observed carefully.

1.7 Ferroptosis

Ferroptosis, another form of regulated cell death which characterized by iron-dependent lipid peroxidation. It has recently gained attention as a potential therapeutic target in cancer. Unlike apoptosis, ferroptosis operates through distinct molecular mechanisms, involving the accumulation of reactive oxygen species (ROS) and lipid peroxides that lead to cell death.[12]

Research suggests that certain cancer cells are particularly vulnerable to ferroptotic stimuli, thus presenting an opportunity for targeted intervention.

1.7.1 Role of Ferroptosis in Ovarian Cancer Treatment

Ovarian cancer cells often have higher iron levels compared to normal cells, which makes them more susceptible to ferroptosis induction since elevated iron levels in cells promotes ferroptosis.

By increasing iron levels or inhibiting iron export can ferroptosis in ovarian cancer cells can be induced. Compounds that trigger ferroptosis, such as erastin and altretamine, show anti-cancer behavior and effectively kill ovarian cancer cells in vitro and in animal models. Thus, by combining ferroptosis inducers with other therapies like SCD1 inhibitors ,the anti-cancer effects can be further enhanced. Key regulators of ferroptosis like SLC7A11, GPX4, and p53 are often altered in ovarian tumors, making them more vulnerable to ferroptosis induction and providing opportunities for selective ferroptosis-based therapies.

Ovarian cancer frequently develops resistance to standard chemotherapies. However, inducing ferroptosis has been demonstrated to reverse this drug resistance, as ferroptosis can overcome drug resistance in ovarian cancer, thus providing a novel approach to treat refractory ovarian tumors.

Key advantages of targeting ferroptosis for cancer treatment:

1) Ferroptosis can overcome drug resistance. Thus cancer cells which maybe resistant to apoptosis-inducing therapies may still be susceptible to ferroptosis inducers, allowing a new approach to reverse drug resistance.

2) Aggressive cancer phenotypes like mesenchymal and dedifferentiated cancer cells, which are often resistant to traditional therapies are easily venerable to ferroptosis.

3) Through combination of ferroptosis inducers with chemotherapy, targeted therapy, or immunotherapy , positive results in preclinical studies for anti-tumor effects has been observed

1.7.2 Limitations

1) Research shows that the specificity and optimal dose of ferroptosis inducers needs to be studied more to prevent damaged to normal cells Achieving the right balance between inducing ferroptosis in cancer cells while avoiding toxicity to healthy cells is an important challenge.

2) The heterogeneity and replasticity of tumor cells affect their sensitivity to ferroptosis inducers , thus affecting the development of effective therapies as cancer cells can have varying susceptibility to ferroptosis

   Reliable ways to monitor ferroptosis induction in patients are still being developed.

Ferroptosis represents a novel form of regulated cell death that is linked to reactive oxygen species accumulation, lipid peroxidation, and the metabolic disruption of iron homeostasis. Currently, studies of ferroptosis are still in their initial stage, though some mechanisms of ferroptosis have been verified, especially in regard to the contributions of iron overload, reactive oxygen species accumulation and lipid peroxidation.[12] However, no specific biomarkers for ferroptosis (like caspase activation for apoptosis) have been identified till date. Thus, identifying specific markers is of great value for the further study of ferroptosis. The available studies on ferroptosis have been based on 2D cell culture and mouse models. Compared with 2D cultures, the application of 3D cultures more accurately reflects the conditions in vivo such as cell–cell interactions and the nature of the extracellular matrix.

Increasing evidence shows that ferroptosis is particulary closely related with cancer suppression. Certain ferroptosis-based therapies have exhibited remarkable potential in anticancer treatments. However, some questions remain: (1) Application of GPX4 inhibitors can be used in cancer therapies based on ferroptosis, but GPX4 inhibition may have side effects on the development of the nervous system. Thus, one of the challenges is to define the physiological roles of ferroptosis. (2) Since the occurrence of ferroptosis is cell-dependent, ferroptosis-based cancer treatments do not apply to all cancer types or even to different clinical stages of the same type of cancer. (3) Ferroptosis is accepted to be immunogenic because it is reported that ferroptosis has positive interactions with immunotherapy in vivo. However, it is not clear that whether ferroptosis has direct immunogenicity.[13][14][15]

Iron-dependent ferroptosis in cancer cells may be triggered by erastin, SAS, or glutamate, inhibiting the system mediated by cystine uptake. RSL3 initiates an analogous iron- and ROS-dependent ferroptotic death program without inhibiting cystine uptake. Importantly, this could enable RSL3 to induce ferroptosis in cells or conditions where survival is not reliant on cystine uptake.

Several substances exhibiting RSL behavior include PEITC, oncrasin, and piperlongumine (Guo et al., 2008; Raj et al., 2011; Trachootham et al., 2006). These substances induce apoptosis features such as caspase activation and mitochondrial cytochrome c release, which are not observed in cancer cells undergoing ferroptosis. Apoptosis is highly resistant in certain tumor cells (Ni Chonghaile et al., 2011).[16] Consequently, drugs capable of causing non-apoptotic death, such as lanperisone, erastin, and RSL3, may demonstrate a distinct spectrum of clinical efficacy.

The intensive study of mechanisms underlying ferroptosis is of vital significance in mapping its role in all kinds of related carcinomas with compelling evidence denoting a close association between ovarian cancer and ferroptosis. As we now know that docetaxel and PARP inhibitors synergize with ferroptosis inducers in OVARIAN CANCER, studies with respect to ferroptosis and other classical drugs such as cyclophosphamide and vincristine in OVARIAN CANCER still await further elucidation.[17] In addition, it has been reported that mature drugs including sulfasalazine could induce ferroptosis in cancer entities like breast cancer and head and neck cancer, their clinical use in OVARIAN CANCER remains poor. Therefore, the expanding clinical usage of those mature drugs towards other malignancies should be considered. Although steady progress has been achieved in terms of ferroptosis, mechanisms underlying the three predominant protection pathways need to be improved and whether there are some other important mechanisms is yet intangible. Conspicuous exploration of ferroptosis and its regulators may provide potential breakthrough points on anti-OVARIAN CANCER therapies

Aberrant regulation of cell death programs plays a pivotal role in tumorigenesis and tumor development. As we know, resistance to apoptosis is a hallmark of cancer (Frank et al., 2019; Koren and Fuchs, 2022)[18][19]. As a newly identified type of regulated cell death (RCD), ferroptosis attracted increasing attention for its critical role in numerous diseases including cancer. Ovarian cancer is a highly heterogeneous tumor with substantial somatic mutations,

accounting for the highest mortality rate among all gynecological malignancies (Yang et al., 2017). ferroptosis inducer erastin synergistically induced ovarian cancer cell death with cisplatin which manifested that targeting ferroptosis in ovarian caner offers therapeutic perspectives (Cheng et al., 2021). the most recent studies discovered that the anti-tumor function of the immune system may associated with ferroptosis. In a recent research aimed at exploring whether ferroptotic cancer cells are immunogenic demonstrated that early stage ferroptotic cancer cells induced by RSL3 were efficiently engulfed by bone marrow-derived dendritic cells (BMDCs) and were able to promote BMDCs maturation and activation. Moreover, early ferroptotic cancer cells in a tumor prophylactic vaccination model showed effective vaccination activity in immune-competent mice. This study provides evidence to support that ferroptotic cancer cells in the early death stages can acts as effective inducers of an adaptive immune response (Efimova et al., 2020).[20]

Recent research in ovarian cancer treatment regarding overcoming cisplatin resistance, underscores the role of ferroptosis in evolving therapeutic strategies. The role of ferroptosis in ovarian cancer, particularly in the context of drug resistance, represents an emerging field with ongoing research.[21] The induction of ferroptosis by agents such as NL01, Tripterygium Glycosides, and iron nitroprusside is used for targeting chemotherapy resistance mechanisms, which used to interfer with the the essential survival processes of cancer cells, and made them resistant to medications like cisplatin.

The identification of specific molecular pathways and targets offers new avenues for therapeutic intervention and suggests a promising future for Ovarian Cancer treatment. The combination of traditional chemotherapy agents with novel compounds has shown increased effectiveness, indicating a powerful strategy to combat drug resistance. Additionally, the role of biomarkers and the importance of fatty acid desaturases in cancer cells highlight potential targets for personalized therapy.

The findings from these studies provide a foundation for further research and development of targeted therapies. The next steps should focus on translating these insights into clinical practice, conducting comparative studies with existing treatments, and exploring broader implications for other types of cancers. This progression is essential for enhancing the prognosis and treatment outcomes of Ovarian Cancer patients.

Apoptosis, or programmed cell death, is a conserved pathway responsible for the development of the organism during embryogenesis and for maintaining tissue homeostasis. Both excessive cell death and resistance to it may lead to pathological conditions. For example, cancer cells resist undergoing apoptosis, followed by uncontrolled cell proliferation. Cancer cells frequently overexpress several anti-apoptotic molecules which prevent the initiation of the apoptotic pathway. By downregulation of pro-apoptotic molecules cells could avoid programmed cell death. Apoptosis is normally receptor-mediated and activated either via signals from mitochondria (named intrinsic pathway) or through extracellular signals (called extrinsic pathway)

Initiating apoptosis in cancer cells is essential, but it requires a number of intricate routes and regulators. It has been observed that cancer cells take use of this procedure to avoid cell death. Even though there are a lot of known targets for ovarian cancer, researchers are still looking for a medication with a strong ability to encourage cancer cell death. Consequently, the goal of several investigations has been to reverse anti-apoptotic processes. Currently, these strategies consist of (i) directly targeting molecules that are modified in cancer cells as part of the apoptotic pathway; (ii) indirectly targeting molecules that result in the inhibition of the apoptotic process; and (iii) targeting cell signalling pathways that enable cancer cells to evade death.

It has been discovered that medications including TW-37, Navitoclax, birinapant, and DEBIO-1143 are effective at mimicking or targeting proteins linked to apoptosis. DPI, PRIMA-1, and INCB054329 are examples of indirect apoptosis inhibitors that demonstrated potential in early ovarian cancer cell tests. Blocking cell signalling pathways such JAK/STAT3, Wnt/β-Catenin, MET/Hepatocyte growth factor (HGF), MAPK/ERK, and PI3K/AKT/mTOR can also cause apoptosis. Numerous research endeavours have examined small chemical inhibitors, including HO3867, Sinomenine, SU11274, BMS-777607, Mab216, and Delphinidin, that function on these signalling pathways.[22]

Ovarian cancer cells can be successfully targeted directly or indirectly to induce apoptosis. Bcl-mediated apoptosis involves numerous interdependent components to induce cell death and is one of the most extensively researched cellular pathways. Clinical trials have shown a remarkable response to BH3 mimetics. Crucially, overexpressing the pertinent apoptotic molecules in the tumours may enhance the effectiveness of these treatments even more. The most often used routes in the treatment of ovarian cancer are those involving p53 and PARP, which have been used as markers for platinum-resistant illness.

In the cellular signaling pathway, there are multiple components that have not been fully understood. Most of these components are involved in more than one cellular process that could either be essential for apoptosis or lead to a different form of cell death. Thus, inhibitors of cellular pathways can cause non-specific targeting of one or many molecules, ultimately causing apoptosis via another mechanism. For example, crizotinib was introduced as a MET inhibitor to target MET/HGF pathway.[23] However, upon further investigations, it became apparent that crizotinib is an effective drug for targeting ALK and ROS1 in addition to MET. Since then, it is being tested as a multi-target drug. Therefore, target specificity should be carefully considered in all circumstances.[24][25]

2. OBJECTIVE

The objective of the project includes:-

→ To identify the differentially expressed genes between healthy and diseased samples

→ To identify the co-expressed genes between apoptosis and ferroptosis processes

→ To understand the relationship between ferroptosis and apoptosis in terms of their respective cellular mechanisms

3. MATERIALS AND METHODS

3.1 DATA COLLECTION

The RNA sequencing reads employed for transcriptomic analysis originated from two distinct project studies, one for diseased tissues and the other for healthy tissues.

The retrieval of the RNA sequencing reads for the transcriptomics study in the form of fastq files was done using the Sequence Read Archive (SRA), formerly called as Short Read Archive which is part of the European Nucleotide Archive (ENA). The reads were retrieved from the project ID PRJNA1005317 and PRJNA578440, as a result of the data generated from the respective studies.

Table 3.1 - Details of RNA Sequencing data from ENA SRA for diseased samples

RNA Sequencingg Reads Detail	
Bioproject	PRJNA1005317
Organism	*Homo sapiens*
Assay type	RNA-Seq
Platform	Illumina HiSeq 300
Library layout	Paired
Library selection	DNAse
Library source	Transcriptomic

Table 3.2- Details of RNA Sequencing data from ENA SRA for control samples

RNA Sequencing Reads De	
Bioproject	PRJNA578440
Organism	*Homo sapiens*
Assay type	RNA-Seq
Platform	Illumina HiSeq 300
Library layout	Paired
Library selection	DNAse
Library source	Transcriptomic

The specifics of the RNA sequencing data, encompassing assay type, library layout, selection criteria, as well as the data source, including platform, project ID, and publication details, are comprehensively documented in the accompanying Table.

The reference genome utilized for alignment, provided in Fasta format, and the reference assembly file, structured as a GTF of the primary assembly, were sourced from ENSEMBL.

The entirety of the data analysis pipeline is executed through the graphical user interface (GUI) of Galaxy, accessible via www.usegalaxy.org and www.usegalaxy.eu, supplemented by the implementation of R programming code within the Windows environment.

3.2 TRANSCRIPTOMICS

Transcriptomics encompasses the comprehensive examination of the entire mRNA repertoire within a cell (Kenny et al., 2011). The workflow delineating the procedures undertaken for the transcriptomic investigation is visually depicted in the accompanying Figure.

The typical pathway for transcript analysis involves the following sequential steps:

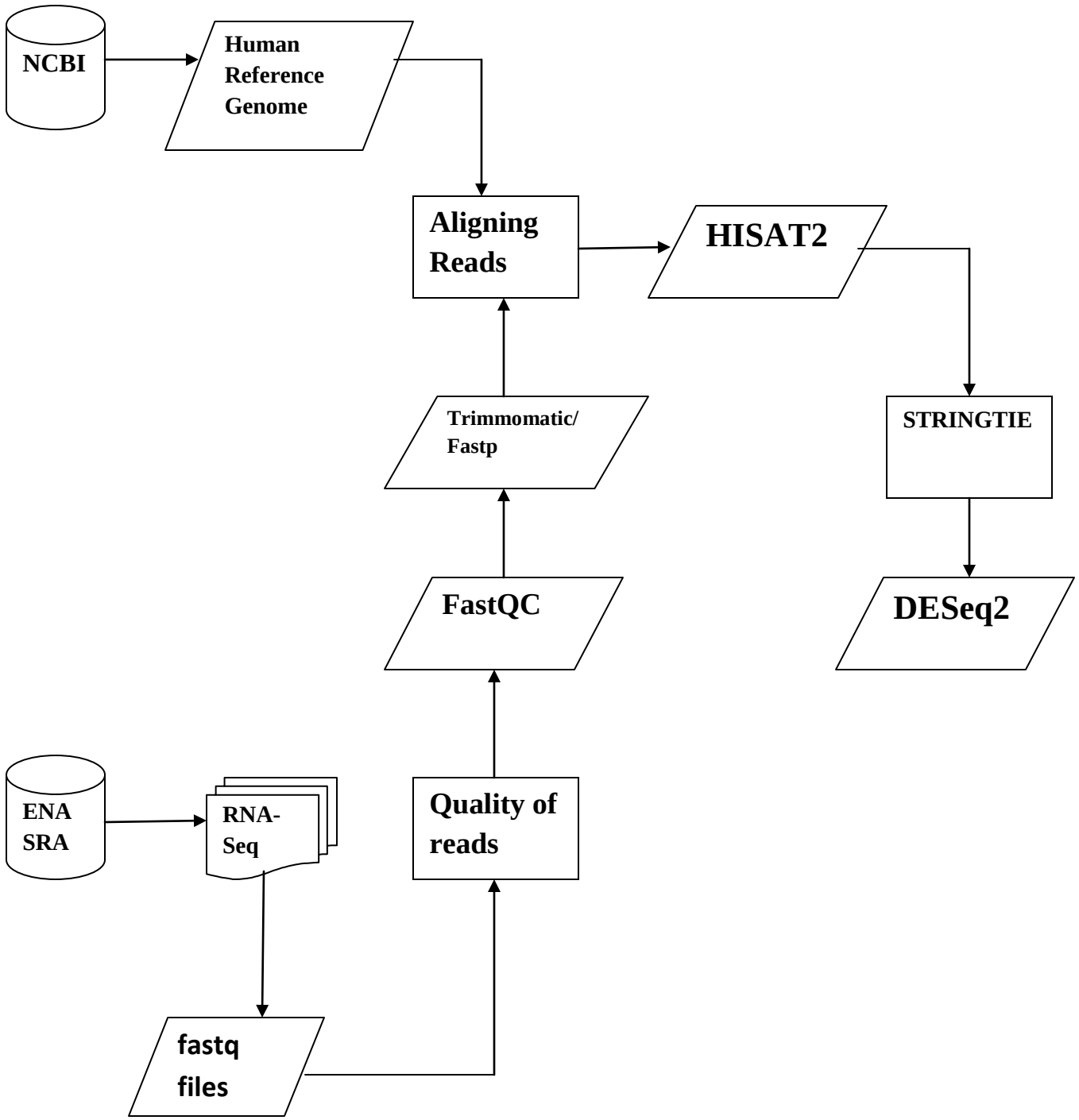
i) Quality Control (QC) assessment of the sequencing reads.

ii) Pre-processing of the reads, including trimming, filtering, and adapter removal.

iii) Alignment of the reads to a reference genome or transcriptome.

iv) Generation of a genome-guided transcriptome assembly to construct full-length transcripts.

v) Quantification of gene expression levels through methods such as read counting or transcript abundance estimation.



(Figure 3.1: Workflow of Transcriptomic analysis)

3.3 QUALITY CONTROL AND PRE-PROCESSING OF READS

Quality control of raw data is contingent upon the specific platforms from which it originates (Trivedi et al., 2014). This process involves scrutinizing the raw data to identify poor-quality reads, considering various parameters such as base sequence quality, sequence content, GC content, sequence duplication, overrepresented sequences, and adapter content. Key considerations in this assessment typically include GC content and sequence duplication.

Multiple tools are available for filtering reads based on quality, including FastQC, PRINSEQ, Trimmomatic, MultiQC, CutAdapt, and FastX. Among these, FastQC (Version 0.73 + galaxy0) is commonly employed for quality assessment due to its user-friendly Java-based graphical interface and rapid analysis capabilities, accepting inputs in FastQ, SAM, and BAM file formats. FastQC presents results in a graphical format, facilitating easier interpretation (Pereira and Sousa, 2020).

Subsequently, reads identified as poor quality can undergo preprocessing using tools such as PRINSEQ and Trimmomatic. These tools are utilized to eliminate adapter sequences attached to reads during sequencing. This preprocessing step enhances read quality, enabling a more effective quality control assessment (Hall, 2015).

3.4 ALIGNMENT READS TO REFERENCE GENOME

The alignment of sequencing reads to a reference genome was performed utilizing the HISAT2 alignment tool, with the reference genome obtained from the ENSEMBL database. Various alignment tools, including HISAT2, BWT, Bowtie2, and TopHat, leverage algorithms such as Burrow Wheeler Transformation, FM Index, Seed-to-Extract algorithm, Hash Table algorithm, and Suffix tree-based methods for this purpose (Tiwary, 2022). HISAT2 utilizes both the BWT and FM index, with the global FM index covering the entire genome and the local FM index focusing on specific genomic regions.

HISAT2 was preferred over alternative tools like TopHat due to its reputation for speed, sensitivity, and efficiency, particularly in handling larger reads. The specific version utilized for this analysis was HISAT2 (Version 2.2.1+galaxy0).

For alignment, HISAT2 was provided with the reference genome and paired-end FastQ files as input, incorporating unstranded strand information and default settings for advanced options within the Galaxy platform. The tool provides output in the form of alignment scores along with BAM files, facilitating downstream processing of the aligned reads.

3.5 GENOME-GUIDED TRANSCRIPTOME ASSEMBLY

The genome-guided transcriptome assembly process yields short transcripts, which may represent either novel entities or those aligned to the reference genome provided in GTF format. Input data for this assembly typically comprises SAM or BAM files, the latter being sorted using Samtools sort (Version 2.0.3) to organize reads based on genomic positions (Biology, n.d.).

Various tools, including StringTie, Cufflinks (Pertea et al., 2016), Trinity, and Velvet+Oasis, are available for transcriptome assembly. Among these, the reference-based assembly tool StringTie (Version 2.1.7+galaxy1) was selected for this analysis.

The input options configured for StringTie include:

- Input options ⟶ Short reads
- Input short mapped reads ⟶ BAM file (Sorted)
- Specify strand information ⟶ Unstranded
- Use a reference file to guide assembly? ⟶ Use reference GTF/GFF
- Reference file ⟶ Use a file from history
- GTF/GFF3 dataset to guide assembly ⟶ GTF file

- Use Reference transcripts only? ⟶ Yes
- Output files for differential expression⟶ DESeq2/edgeR/limma-voom
- Advanced Options: Output gene abundance estimation file? ⟶Yes

Execution of these options generated outputs in the form of a gene abundance file (GTF), assembled transcripts, gene count, and transcript count files. To identify novel transcripts, the option "Use Reference Transcripts only" was set to "NO," enabling the listing of novel transcript sets. Subsequently, HTSeq2 was employed to generate gene counts, and the resulting gene count files were merged into a tabular output using the "Column merge" tool within the Galaxy platform.

3.6 DIFFERENTIAL GENE EXPRESSION ANALYSIS

Following the execution of the specified options, the analysis yielded several outputs, including a gene abundance file (GTF), assembled transcripts, gene count, and transcript count files. To identify novel transcripts, the parameter "Use Reference Transcripts only" was configured to "NO," facilitating the detection of novel transcript sets.

3.7 DOWNSTREAM PROCESSING OF DEGs

The DEGs identified from the reference transcript-based assembly were analysed for the functional annotation with the gene ontology study and the pathways in which the genes are involved. The gene IDs were identified for the corresponding DEGs using Ensembl Biomart using the following steps within the server,

CHOOSE DATABASE ⟶ Ensembl Genes 106

CHOOSE DATASET ⟶ Human genes (GRCh38.13)

Filters:

GENE ⟶ Input external references ID list [Max 500 advised]

Paste or upload the DEGs identified

Attributes:

GENE ⟶ Choose Gene Stable ID, Gene Name, Gene type

Result

This is done by using the STRING, Version 11.5 (https://string-db.org) for the functional annotation and interaction analysis and GeneMania (https://genemania.org/) to find the interaction of the input DEGs and the other genes.

The STRING database aims to collect the functional interaction between the expressed proteins which is the result of the translation of the DEGs identified (Szklarczyk, et al. 2017). The interactions are derived from Genomic content prediction, High-throughput lab experiments, Co-Expression, Automated data mining, and Knowledge of databases.

3.8 PROTEIN-PROTEIN INTERACTION

Protein-Protein Interactions (PPIs) are physical contacts of high specificity established between two or more protein molecules as a result of biochemical events. Densely connected subgraphs in PPI networks often represent protein complexes that function together in biological processes. Protein hubs with a large number of interactions in PPI networks tend to include important proteins like enzymes, transcription factors, and intrinsically disordered proteins. Orienting PPI networks to determine the direction of signal flow improves the ability to predict protein function, and identify cancer driver genes and drug targets.

PPI interactions between the gene network reconstruction results were carried out with the help of STRING database.

The **STRING database** (https://string-db.org/), which stands for Search Tool for the Retrieval of Interacting Genes/Proteins, is a resource dedicated to functional associations between proteins. It provides protein-protein interaction networks and covers a vast number of organisms, with the latest version 10.0 expanding its coverage to 2031 organisms. Users can access STRING through its website, a REST-based API, or by downloading wholesale data. The database offers various features like network visualization, interaction evidence inspection, score adjustments, and statistical enrichments. In addition to this, access to STRING from the R/Bioconductor environment is facilitated through the STRINGdb package, allowing users to interact with the database programmatically. The database's user interface allows for protein searches, network exploration, and advanced functionalities like clustering, rearranging networks, and detecting enrichments related to human diseases and tissues. STRING also collaborates with partner databases for additional annotations and enrichments, enhancing its utility for biological research and analysis.

These interactions were then visualized and analyzed using the plugin NetworkAnalyzer (v4.5.0) in Cytoscape.

Cytoscape is an open-source bioinformatics software platform used to integrate gene expression profiles and other state data with molecular interaction networks to visualize them. Its function includes network visualization, integration with diverse attribute data, and support for various graph theory use cases and for complicated network analysis. The fundamental function of Cytoscape is to depict molecular species as nodes in a network graph and their interactions as links. Essential tools present in Cytoscape are tools for data integration, visualization, selection, and filtering are included in the software, and plug-ins allow for the incorporation of external methods.

3.9 GENE NETWORK RE-CONSTRUCTION

Gene network reconstruction is the process of studying the interactions between the genes based on gene expression data. This provides information regarding the underlying regulatory mechanisms within a biological system and their relations with each other.

Here, the Gene Network is reconstructed through:-

Pearson's Correlation Test:- In order to identify possible regulatory links between genes, gene expression data is analyzed as part of the process of gene network reconstruction using Pearson's correlation test. This method begins by gathering the gene expression levels in various samples. The degree and direction of each gene pair's linear link are then measured using Pearson's correlation coefficient. Genes that show statistically significant associations are chosen according to preset thresholds that suggest possible interactions. Genes are the nodes and these strong correlations are the edges of a network that forms an extensive map of regulatory linkages inside the biological system.

3.10 GENE CO-EXPRESSION NETWORK

A gene co-expression network was constructed among differentially expressed genes associated with ferroptosis and apoptosis, employing the PSYCH package within the R programming environment. This network comprised of both positively and negatively co-expressed correlation networks, which were further visualized and analyzed using Cytoscape.

Genes exhibiting higher interactions (degree) within the network were comparedwith protein-protein interaction (PPI) results to identify potential key candidate genes and their interrelationships.

3.11 FUNCTIONAL ENRICHMENT ANALYSIS

A functional enrichment analysis study can be used to determine the biological activities linked to a set of genes and determine whether or not they are functionally connected. The network as a whole or a specific subset of an interest cluster can be the subject of this investigation.

DAVID(https://david.ncifcrf.gov/summary.jsp)is used to find the carcinogenic-associated pathway of the differentially expressed apoptosis and ferroptosis genes

DAVID (Database for Annotation, Visualization and Integrated Discovery) is a bioinformatics tool that provides a comprehensive set of functional annotation tools for researchers to understand the biological meaning behind large lists of genes. It allows users to submit gene lists, convert gene identifiers, and perform various analyses to gain insights into the functional enrichment of their genes of interest

4. REULTS AND DISCUSSION

4.1 DATA COLLECTION

The raw data in the form of fastq files collected from the ENA SRA for the transcriptomic study were filtered to choose only from RNA-seq assay cell types. The severity of the disease and the control datasets were identified using the NCBI Run Selector. The severely diseased cancerous cells based on the presence of BRCA1/2 mutations were chosen with the control and diseased samples.

(Table 4.1 list of the SRA Run IDs of the control and the diseased samples.)

Diseased Samples	Control Samples
SRR25637180	SRR10313349
SRR25637181	SRR10313350
SRR25637182	SRR10313351
SRR25637185	SRR10313352
SRR25637188	SRR10313353
SRR25637205	SRR10313354
SRR25637208	SRR10313355
SRR25637216	SRR10313354
SRR25637217	SRR10313355
SRR25637220	SRR10313356
SRR25637221	SRR10313357
SRR25637222	SRR10313358
SRR25637223	
SRR25637224	
SRR25637225	

4.2 TRANSCRIPTOMICS

QUALITY CONTROL OF READS

The FastQC tool was employed to filter the reads according to quality parameters, revealing suboptimal quality across various metrics such as per tile sequence quality, per sequence GC content, sequence duplication levels, and the presence of overrepresented sequences. Subsequently, within the dataset comprising RNA-seq cell types, the number of control samples was reduced from 12 to 6, while the diseased samples decreased from 15 to 7, through the filtration process.

(Table 4.2 list out the filtered samples after quality check by FastQC tool.)

Diseased samples	**Control Samples**
SRR25637181	SRR10313349
SRR25637182	SRR10313350
SRR25637208	SRR10313351
SRR25637216	SRR10313352
SRR25637217	SRR10313353
SRR25637220	SRR10313354
SRR25637224	

4.3 ASSEMBLY OF READS TO THE REFERENCE GENOME

The good quality reads were aligned with the reference genome where the alignment score was found to be greater than 90% ,mostly between the ranges of 95– 98% for most of the cases. However, few reads showed alignment score between 40-60%. Such reads were discarded and the number of diseased samples reduced from 10 to 5, while the control samples

remained the same number. The reads showing more alignment scores are subjected to the genome assembly and used to find the differentially expressed genes (DEGs).

(Table 4.3 list out the alignment scores in percentage for the diseased and control samples respectively.)

Diseased ID	HISAT Score %	Control ID	HISAT Score %
SRR25637181	97.23%	SRR10313349	96.30%
SRR25637182	97.56%	SRR10313350	96.29%
SRR25637208	42.36%	SRR10313351	96.48%
SRR25637216	96.88%	SRR10313352	96.26%
SRR25637217	97.48%	SRR10313353	96.42%
SRR25637220	97.86%	SRR10313354	96.54%
SRR25637224	56.23%		

4.4 GENOME GUIDED TRANSCRIPTOME ASSEMBLY

The StringTie tool serves to furnish assembled transcript data, consisting of lists of transcripts and their corresponding exons. This output provides crucial quantitative metrics such as coverage, Fragments Per Kilobase of transcript per Million mapped reads (FPKM), and Transcripts Per Million (TPM) values for each gene, all standardized to a score of 1000. This process is iterated for both reference-based transcript assembly and de novo assembly approaches.

In the de novo assembly output, genes are delineated in the format "STRG", with each transcript categorized as either novel or known based on accompanying notes or comments within the transcript list.

The assembled transcript file required for furthur analysis, provides information on transcript regions and their start and end sites, alongside alignment scores relative to the reference Gene Transfer Format (GTF) file. This facilitates comparative analysis and validation against known genomic features.

Furthermore, the gene abundance file, in conjunction with gene count and transcript count files, proves instrumental in identifying novel transcripts and elucidating their expression patterns within the analyzed dataset.

4.5 DIFFERENTIAL GENE EXPRESSION ANALYSIS

The Bioconductor R package was installed to R and the biomart is used to install the plots required for the analysis. The gene count file which was merged contains nearly 7,862 genes with the reference GTF and 59,251 genes without the reference.

(Table 4.4 : Narrowed the gene count based on the significant value (p value))

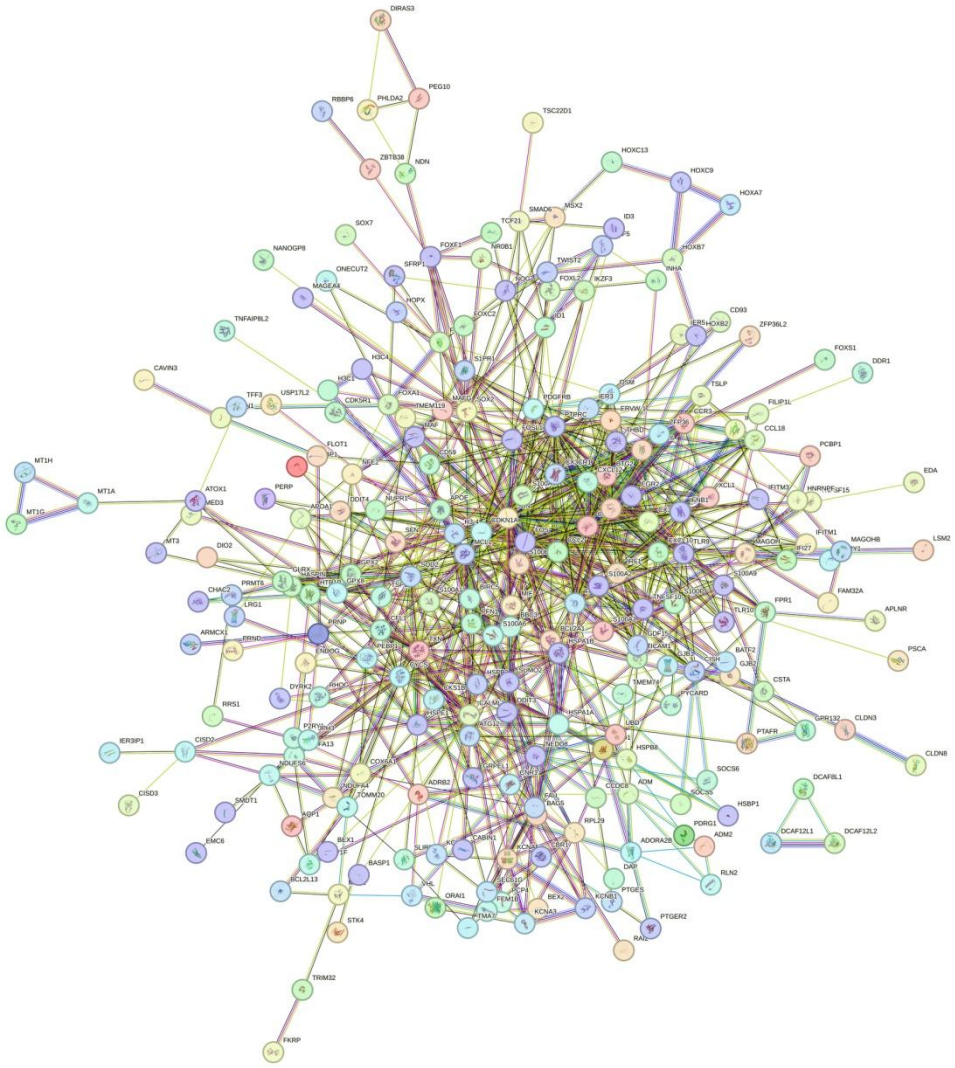
Total genes	7,862
P value <0.05	8002
P value < 0.01	6577

The corresponding genes IDs for the observed DEGs were identified using the ensembl biomart where the protein coding genes were identified. The protein coding genes were chosen to proceeds with the functional annotation and gene ontology studies.

4.6 PROTEIN-PROTEIN INTERACTIONS

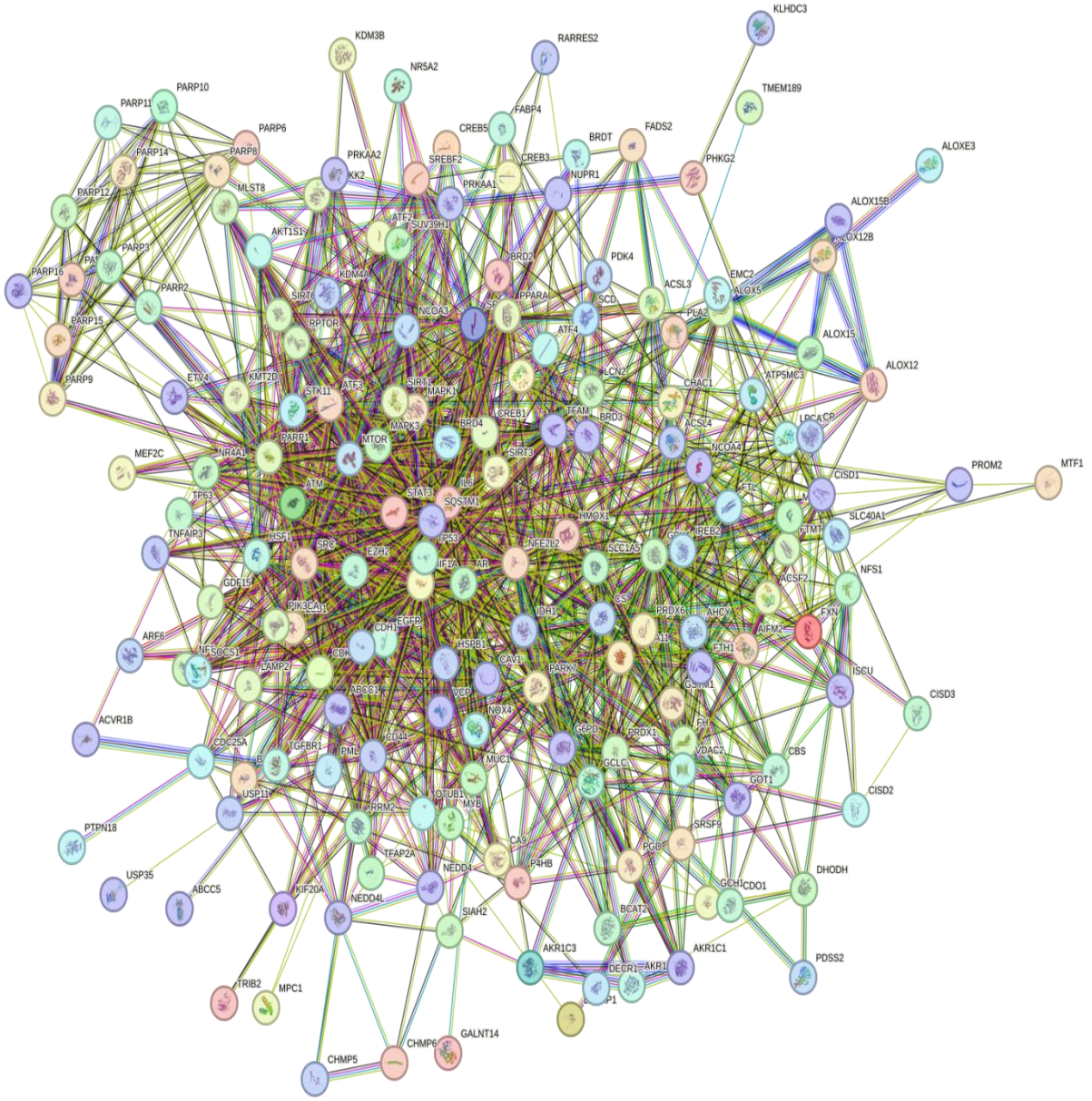
To identify the key genes and their interactions the PPI network was reconstructed using the differentially expressed genes of Ferroptosis and Apoptosis. A total of 289 genes were filtered out from the 7862 genes by using various filters of log2 and p-value

Occurrence of 236 nodes and 886 interactions were found for differentially expressed genes

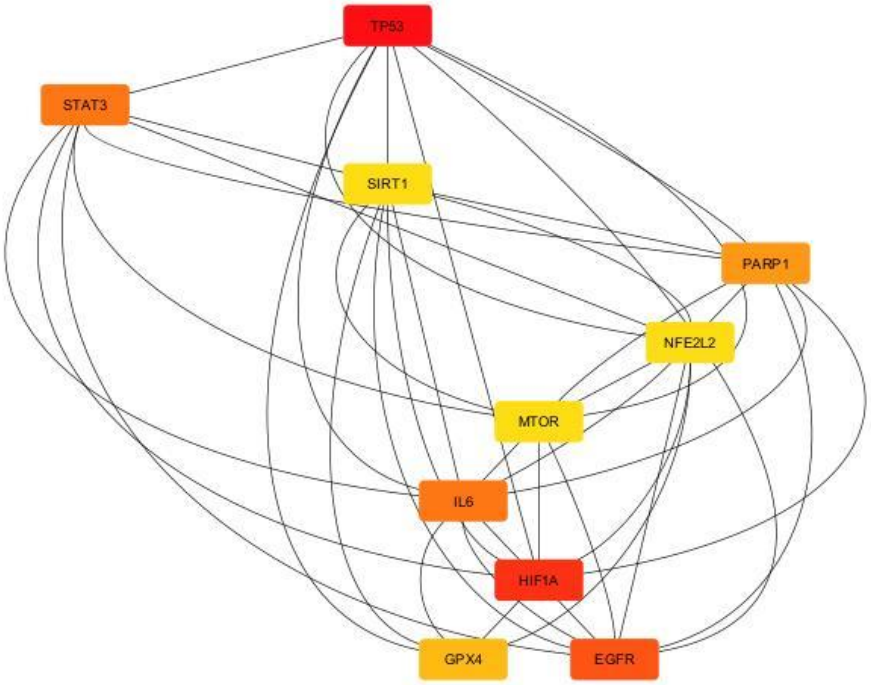


(Figure 4.1: PPI network of Diferentially annotated genes)

Ocurrence of 174 nodes and 1469 interactions were found for differentially expressed ferroptosis genes.



(Figure 4.2: PPI network of Ferroptosis gene)



(Figure 4.3: Visual representation of Ferroptosis hub genes)

The nodal ranking of the hib genes and their designation is given below. There has been 25 interactions between 12 nodes.

(Table 4.5 : List of Nodes and their rank)

RANK	NODE
1	TP53
2	HIF1A
3	EGFR
4	IL6
4	STAT3
6	PARP1
7	GPX4
8	SIRT1
8	MTOR
8	NFE2L2

4.7 GENE CO-EXPRESSION NETWORK

The gene co-expression network includes co-expression interactions between genes of positive co-expression network and genes of negative co-expression network.

A total of 223 apoptosis genes and a total of 7 ferroptosis genes were in found in positive co-expression network while a total of 143 apoptosis gene and a total of 7 ferroptosis genes were found in negative co-expression network

These interactions were then compared with PPI to identify the candidate genes and interactions.

A total of 4 common ferroptosis hub genes were found from positively and negatively co-expressed network--

- CDKN1A
- GDF15
- CISD2
- NUPR1

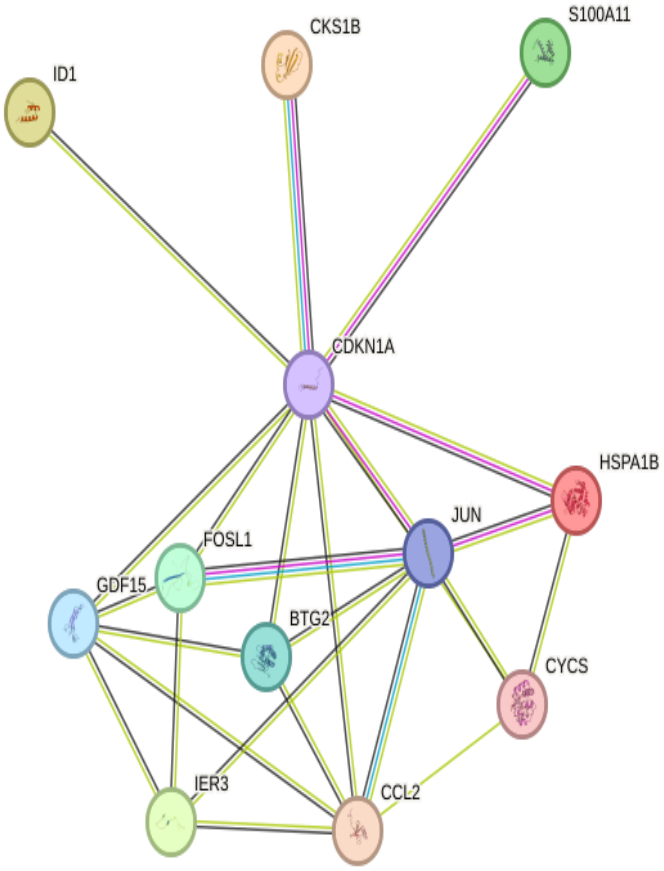
Out of these CDKN1A and GDF15 were chosen because of their high interactions (23 and 10 respectively)

(Table 4.6: List of Ferroptosis and Apoptosis co-expressed genes in Positively correlated network)

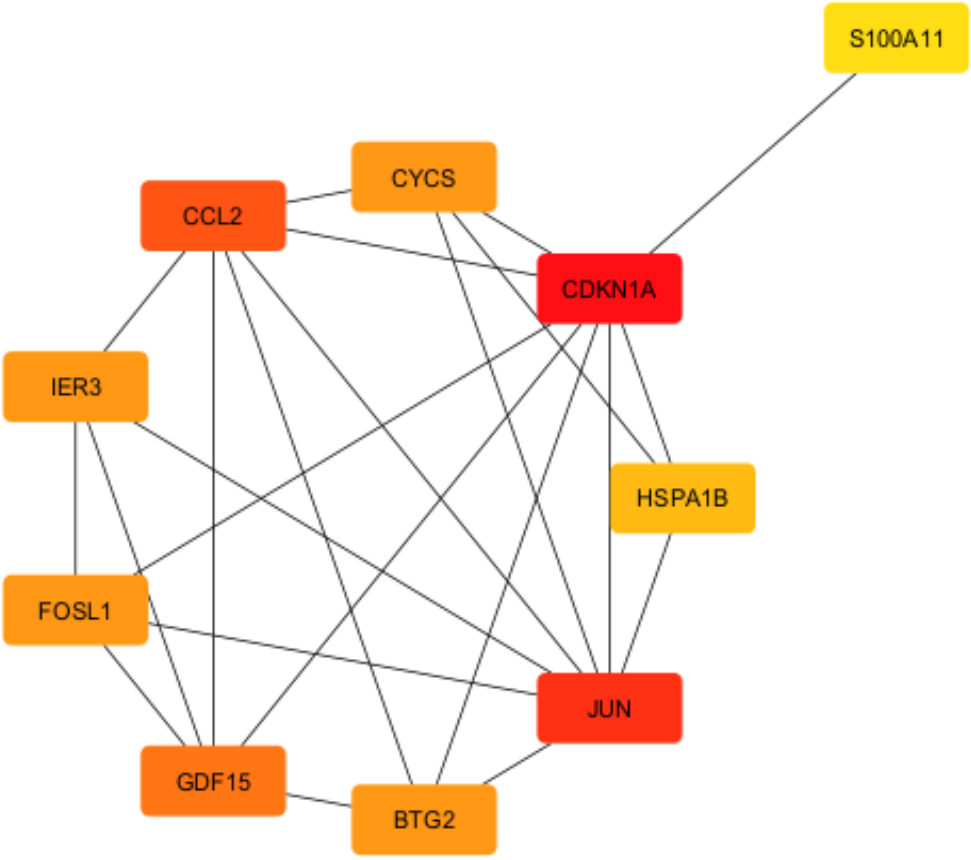
Ferroptosis	**Apoptosis**
CDKN1A	BTG2
GDF15	CCL2
CDKN1A	FOSL1
CDKN1A	HSPA1B
CDKN1A	ID1
CDKN1A	JUN

(Table 4.7 : List of Ferroptosis and Apoptosis co-expressed genes in Negatively correlated network)

Ferroptosis	**Apoptosis**
CDKN1A	CCL2
CDKN1A	CKS1B
CDKN1A	CYCS
GDF15	FOSL1
GDF15	IER3
CDKN1A	S100A11



(Figure:4.4 PPI network of co-expressed ferroptosis and apoptosis genes)



(Figure 4.5: Degree interaction network of co-expressed ferroptosis and apoptosis genes)

(Table 4.8 : List of Nodes and their rank)

Rank	Node	Type of gene
1	CDKN1A	Ferroptosis
2	JUN	Apoptosis
3	CCL2	Apoptosis
4	GDF15	Ferroptosis
5	BTG2	Apoptosis
5	FOSL1	Apoptosis
5	IER3	Apoptosis
5	CYCS	Apoptosis
9	HSPA1B	Apoptosis
10	S100A11	Apoptosis

4.8 FUNCTIONAL ENRICHMENT ANALYSIS

The corresponding Protein-Protein interactions showed that these genes are associated with the carcinogenic pathway. Genes having higher degree of interaction were considered to be candidate key genes and their interactions with the apoptosis genes were considered as possibilities for the formation of the co-expressed network.

Pathways being regulated by both ferroptosis and apoptosis genes were studied and tabulated in the following table

(Table 4.9 : List of KEGG pathways regulated by both Ferroptosis and Apoptosis co-expressed genes)

TERM	**GENES**
Colorectal cancer	CDKN1A, JUN, CYCS
Small cell lung cancer	CDKN1A, CYCS, CKS1B
Pathways in cancer	CDKN1A, JUN, CYCS, CKS1B
Human T-cell leukemia virus 1 infection	FOSL1, CDKN1A, JUN
Renal cell carcinoma	CDKN1A, JUN
Platinum drug resistance	CDKN1A, CYCS
p53 signaling pathway	CDKN1A, CYCS

The primary function of CDKN1A, sometimes referred to as p21, in cancer prohibition is linked to its position as a tumor suppressor. By inhibiting the cyclin-dependent kinases, it controls the cell cycle at the G1 phase, which is essential for controlling cell proliferation, as a response to DNA damage and cell cycle advancement. The tumor suppressor protein p53,

which causes cell cycle arrest in response to different stress stimuli through CDKN1A, strictly regulates this regulation.

It has been observed that in human fibrosarcoma cells, cystine deprivation-induced ferroptosis is delayed through the mediation of p53 function by CDKN1A. This suggests that CDKN1A plays a role in the way cells react to situations that cause ferroptosis.

JUN is an oncogenic transcription factor controling the expression of genes linked to invasion and proliferation, metabolic reprogramming, and cell cycle progression—all of which promotes cancer development. Aggressive human sarcomas and other cancers have been shown to overexpress the JUN gene, to promote tumor development and spread. c-Jun is significantly overexpressed in a large number of malignancies, which encourages cell division and suppresses apoptosis, which is an indication of cancer.

Apoptosis (a process that eliminates damaged or abnormal cells) is triggered by CYCS (Cytochrome c) release from mitochondria triggers, thus acting as a tumor suppressor mechanism. This function helps to prevent the proliferation of cancer cells. However, dysregulation of cytochrome c can lead to resistance to apoptosis, allowing cancer cells to evade cell death and promote tumor growth.

CKS1B is a member of the conserved cyclin kinase subunit 1 (CKS1) protein family and plays an essential role in cell cyclin and promotes cancer development by enhancing cell growth, invasion, metastasis, and drug resistance. However, CKS1B can be considered a potential therapeutic target for cancer treatment, since targeting CKS1B could help overcome drug resistance and improve treatment outcomes

Studies have indicated that CKS1B functions in conjunction with apoptosis and other types of cell death when ferroptosis is induced. A synergistic connection between ferroptosis and apoptosis was observed in studies on human pancreatic and colorectal cancer cells treated with ferroptotic drugs such as erastin and artesunate in combination with the apoptotic agent TRAIL.

FOSL1 promotes cancer progression by enhancing cell proliferation, epithelial-mesenchymal transition (EMT), migration, invasion, and metastasis. Depletion of FOSL1 in cancer cells with KRAS mutations leads to decreased cell viability, shortened S-phase, increased G2 phase, and induction of apoptosis, suggesting its importance in KRAS-driven cancers

FOSL1 is involved in regulating gene expression related to cell differentiation and tumorigenesis. Studies indicate that downregulation of FOSL1 can serve as a potential treatment option for anticancer therapy, due to its role in promoting cell survival and tumorigenesis

Upregulation of FOSL1 alters the expression of core AP-1 complex members, influencing cellular processes related to apoptosis and cell cycle progression

The interplay between ferroptosis and apoptosis involves the induction of endoplasmic reticulum stress and the upregulation of the pro-apoptotic protein PUMA via the C/EBP-homologous protein (CHOP) pathway in response to ferroptosis inducers like erastin and artesunate

The involvement of the p53-independent CHOP/PUMA axis in response to ferroptosis inducers suggests a key role for FOSL1 in sensitizing cells to TRAIL-induced apoptosis during ferroptosis

The interaction between ferroptosis and apoptosis occurs because of ferroptosis inducers such as erastin and artesunate which causes endoplasmic reticulum stress and upregulates the pro-apoptotic protein PUMA through the C/EBP-homologous protein (CHOP) pathway. The involvement of the p53-independent CHOP/PUMA axis in response to ferroptosis inducers suggests a key role for FOSL1 in sensitizing cells to TRAIL-induced apoptosis during ferroptosis.

5. CONCLUSION

The study of the co-expressed ferroptosis and apoptosis genes resulted in the discovery of four key ferroptosis hub-genes having common regulatory relationship with apoptosis genes. They include

- CDKN1A
- GDF15
- CISD2
- NUPR1

Out of these CDKN1A and GDF15 were chosen because of their high interactions (23 and 10 respectively)

The work involves the identification of the carcinogenic pathway being regulated by both ferroptosis and apoptosis genes including renal cell carcinoma,, platinum drug resistance, colorectal cancer etc. The methodology was done completely using Galaxy GUI web - based server for the data analysis

To sum up, understanding the way ferroptosis and apoptosis interact in ovarian cancer offers important information for the creation of novel treatment strategies. Further research is required to resolve and understand the details about this interaction and translate these findings into clinical practice for more effective treatment of ovarian cancer**.**

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